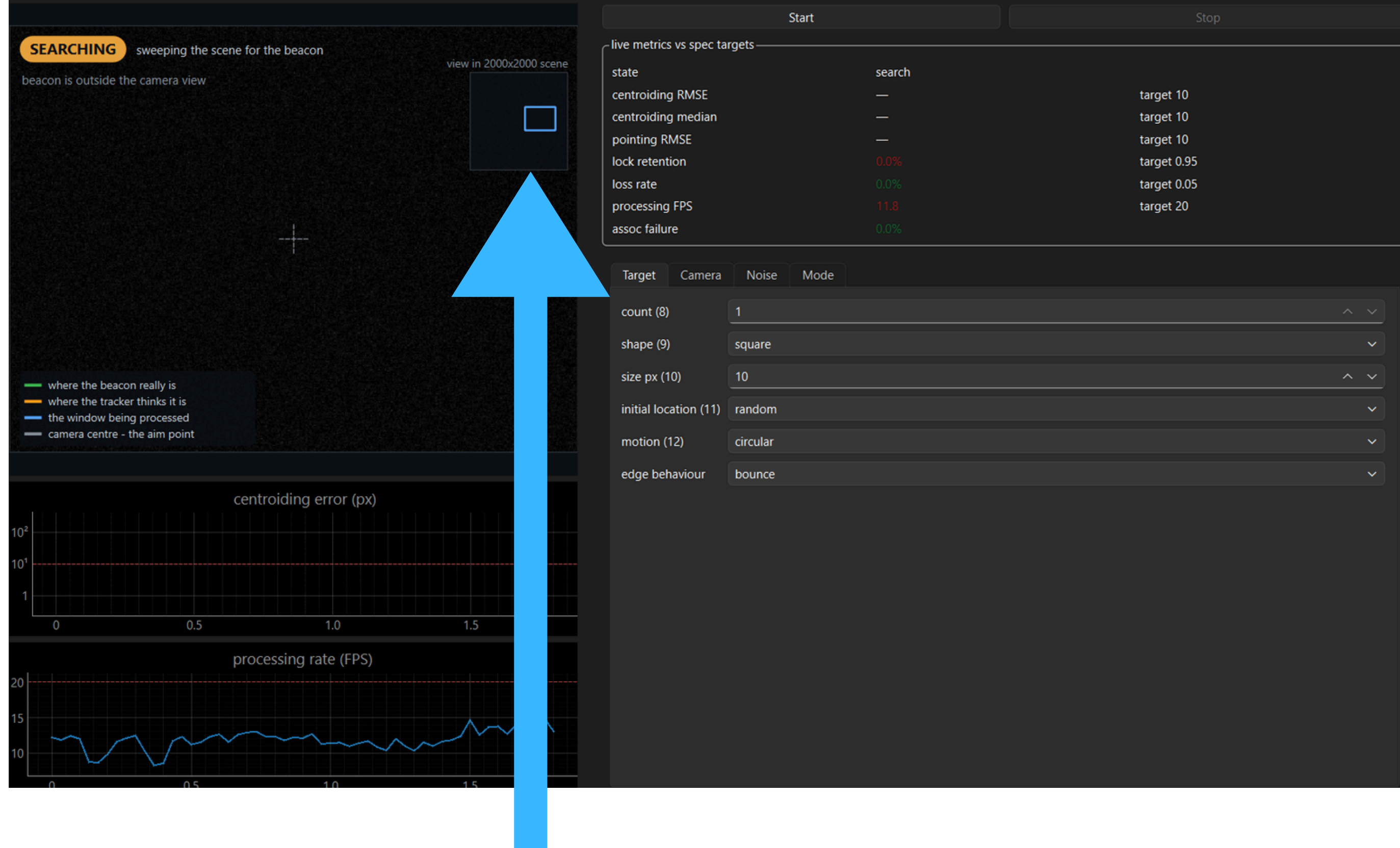
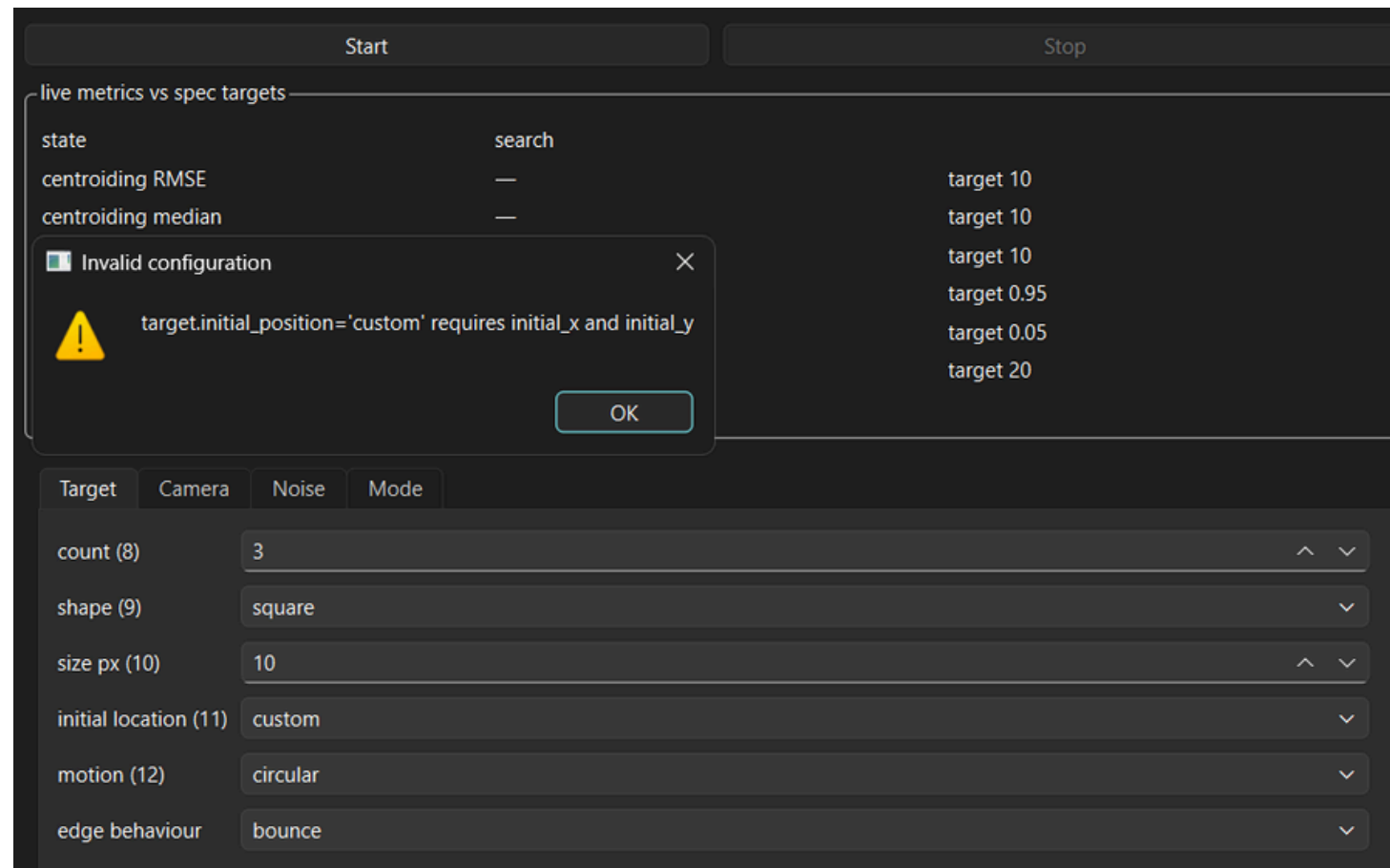
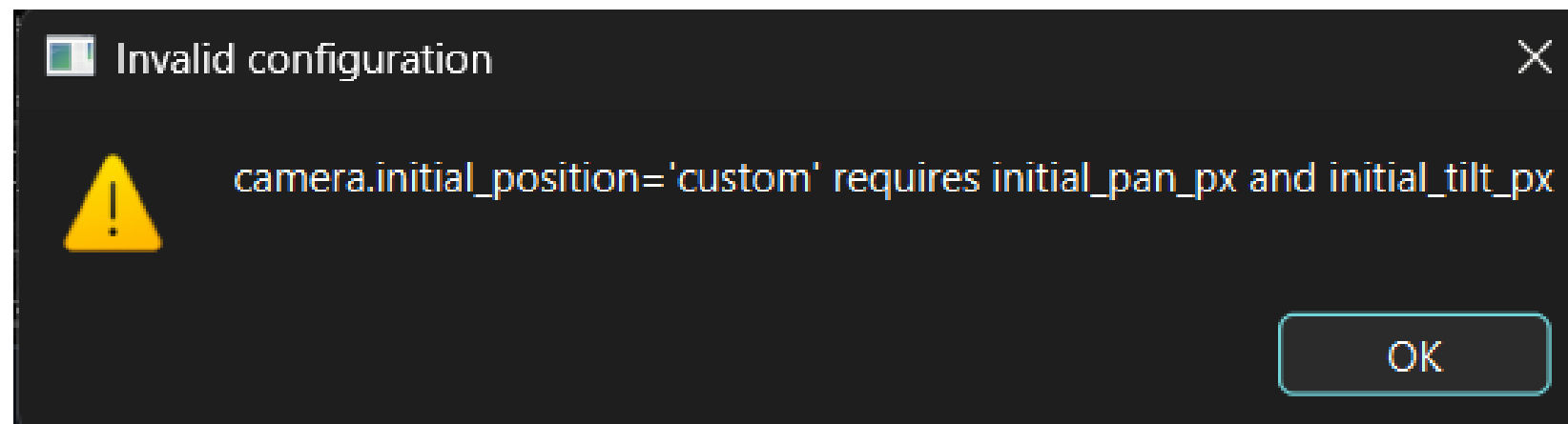




in this window the moving targets which the cam is searching for should be visible as a white dot or something. rn only the camera frame is moving nothing else is happening

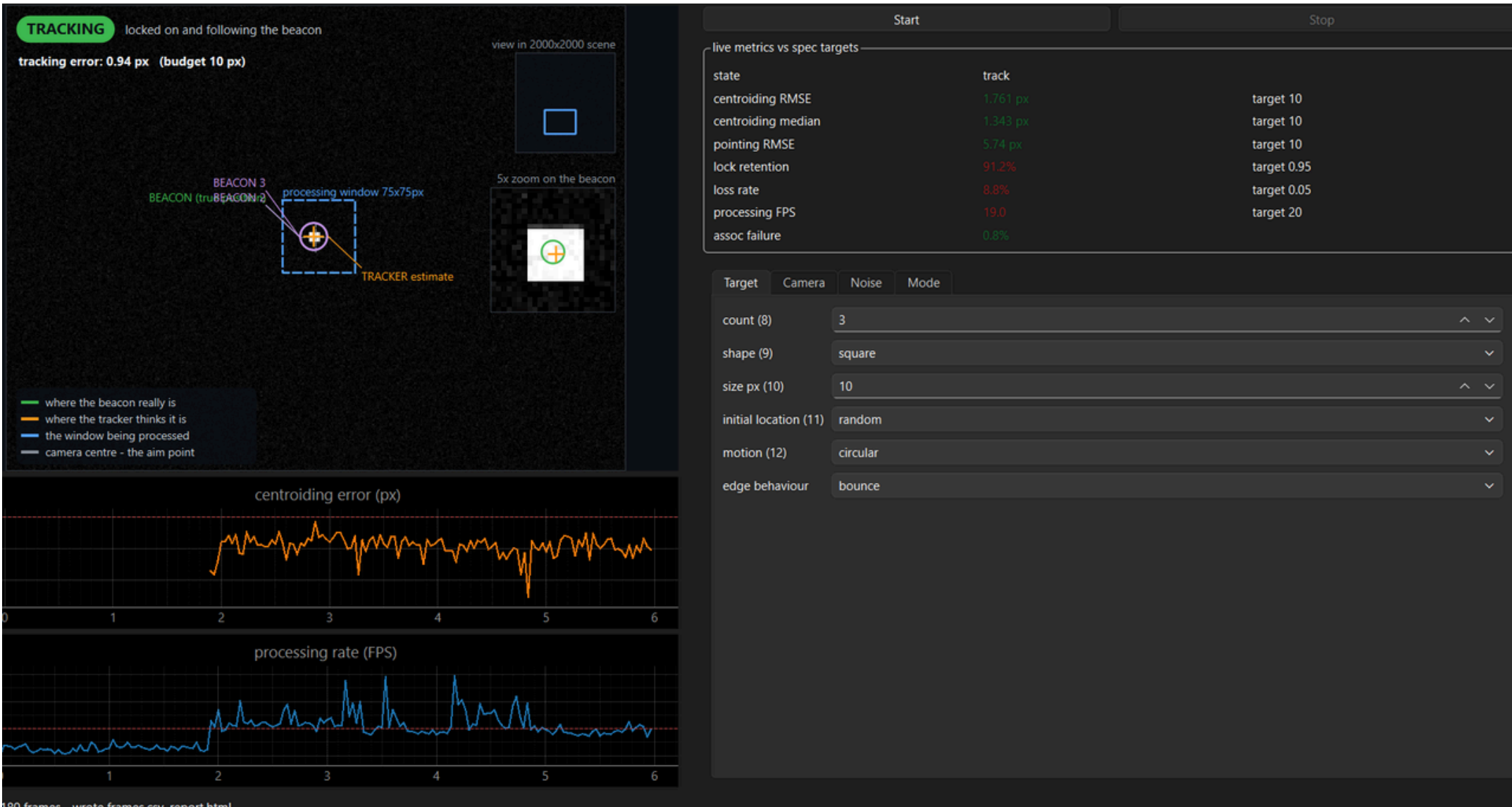


not able to set initial target location

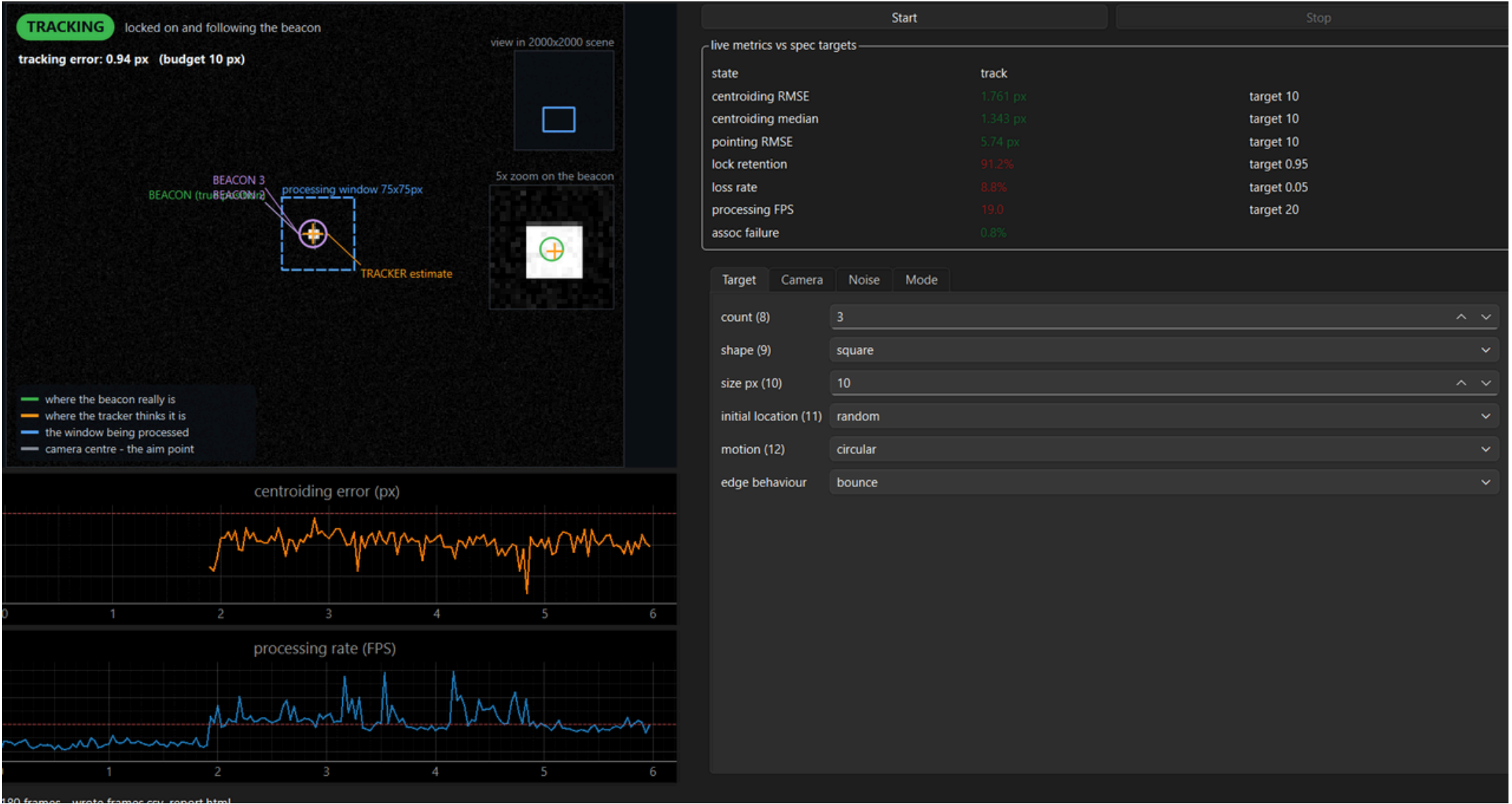


same for camera but its not
mentioned in the doc either so let it
be

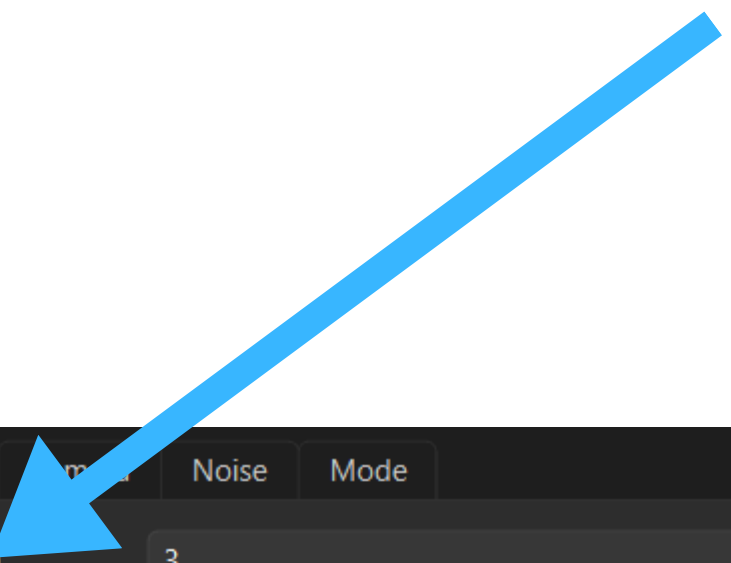
able to change number of targets but not able to visually differentiate them.they are supposed to be dummy targets and our system is supposed to find the most beacon looking target out of all 3



not only are they spawning on the same original beacon they are also following the same path as the og beacon.each target should have its different path.and all the targets should be visible in the minimap



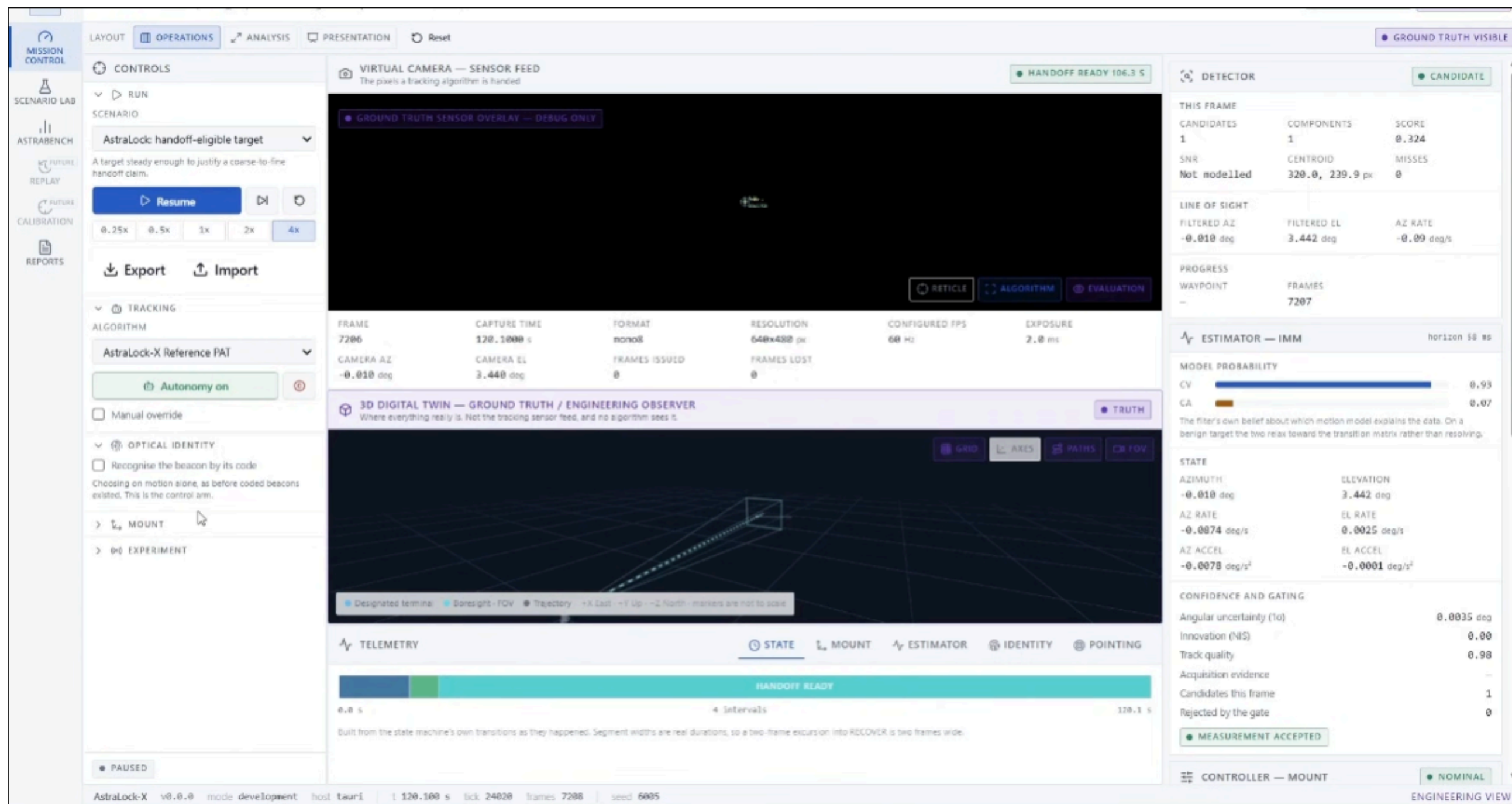
can the targets leave behind a trail so
its visible what motion they are
doing. right now cant differentiate
their motions



Target	Parameter	Noise	Mode
count (8)	3		^ v
shape (9)	square		v
size px (10)	10		^ v
initial location (11)	random		v
motion (12)	circular		v
edge behaviour	bounce		v

the numbers in the brackets are not necessary are they? maybe for demo

instead we could have the max and min threshold of the field be mentioned there



saglyani toch format thevlay 2 screens cha

<https://www.youtube.com/watch?v=VWE3Dq6RUtM>

<https://github.com/Pranavsingh431/AstraLock-X>

ai vaparlay aplya project madhe jevha multiple targets madhun barobar vala beacon shodhayla(can also use yolo to detect the correct target)

Your system first uses normal computer vision to find the possible beacon.

Then, if there are multiple possible targets or the result is unclear, the CNN (AI) helps decide which candidate is more likely to be the real beacon.

So:

Camera image → Computer Vision finds candidates → AI/CNN helps choose the correct one → IWCoG finds the exact centre → Kalman tracks it → PID moves the camera.

- Display tracking performance and statistics in real-time

Deliverables

Each participating team shall submit the following mandatory deliverables:

Software Application

A standalone executable application implementing the complete virtual camera tracking system. The application shall provide all the mandatory functions and features as described above.

Source Code

Complete source code with proper documentation. The code shall be modular and adequately commented.

Technical Report

The technical report (about 10-15 pages) containing problem understanding, system architecture, description of software modules, tracking methods, AI methods (if used), test methodology, performance analysis and future improvements shall be submitted.

User Manual

The user manual with the description of installation of software, application operation, parameter configuration, GUI description, etc. shall be submitted. A 3–5 minutes video may also be provided as an optional deliverable for demonstration of the application.

Performance Log

The software should be capable of automatically generating a performance report containing simulation duration, FPS, acquisition time, average and maximum tracking error, lock retention rate, processing time, etc.

Evaluation Method and Criteria

The solutions developed by participating teams will be evaluated using multi-layered evaluation method. The following table describes stages of evaluation, their weightage and methods.

Expected Solution:

Participants shall develop an AI-assisted camera tracking system capable of automatically detecting and continuously tracking a moving optical beacon in a simulated video stream while controlling a virtual pan-tilt camera.

at the same time ithe lihilay ki “AI methods(if used)”

mhanje compulsory nahi vatta ai

they also want the stats to be real time(not achieved)and downloadable(achieved)

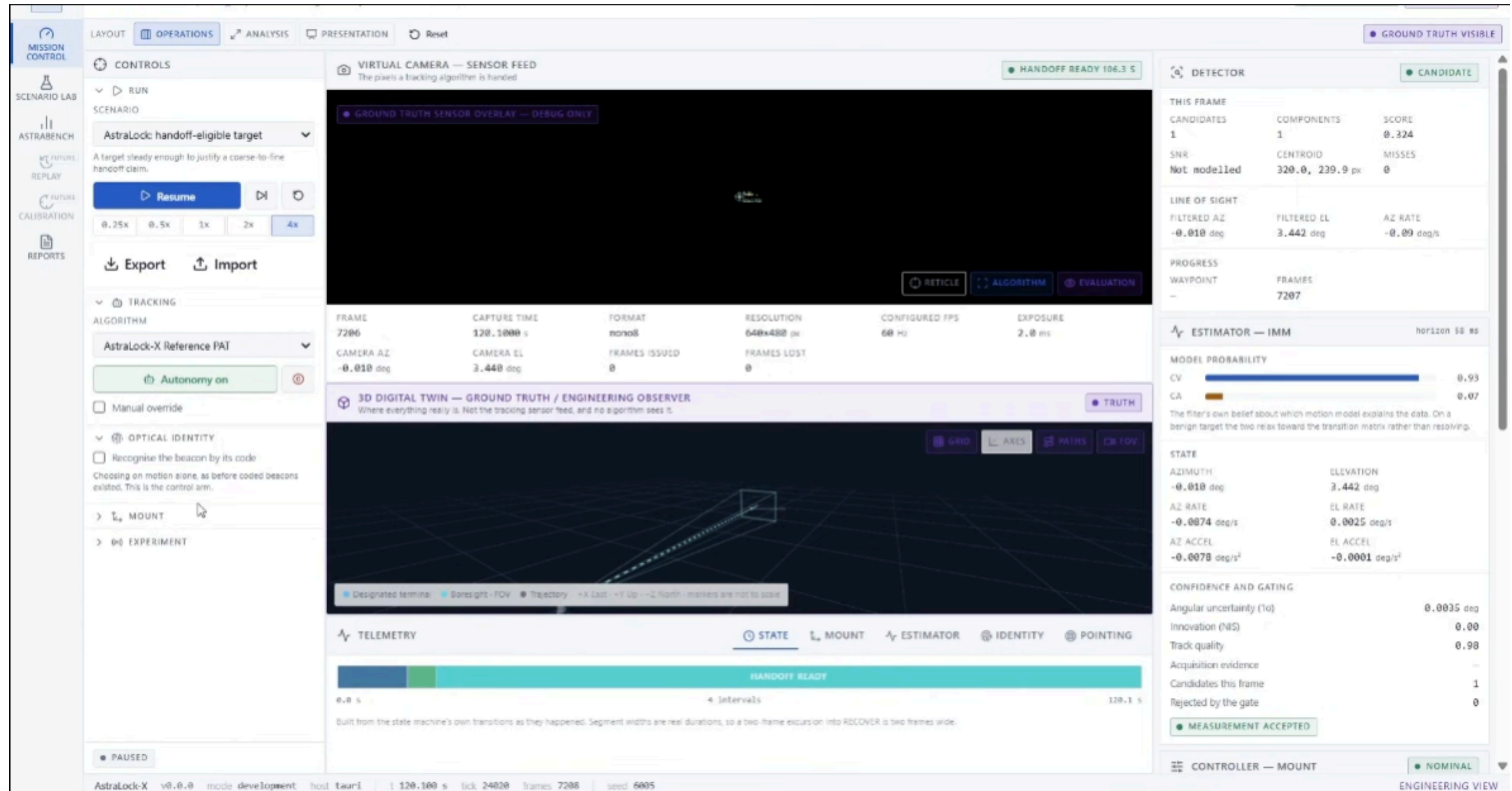
mhanje simulation chalta chalta at the same time screen var dislay pahijet i think

Teams present their approach, methods, architecture, design, etc to the evaluators.

1. Understanding of the problem
2. System architecture and software design
3. Selection of Algorithms
4. AI and computer vision
5. Innovation and Novelty
6. Technical Documentation and Presentation
7. Technical Discussion and Q&A

20

kahitari navin karnyasathi i think fov wali screen 3d
karun camera drag karun dakhvaycha in any line of
axis



git repo link public after sept 30 for both ps

Description:

Unlike conventional radio-frequency systems, FSOC relies on a highly directional optical beam. Even a small angular error can prevent successful communication. Before fine pointing mechanism can take over, a coarse alignment stage must:

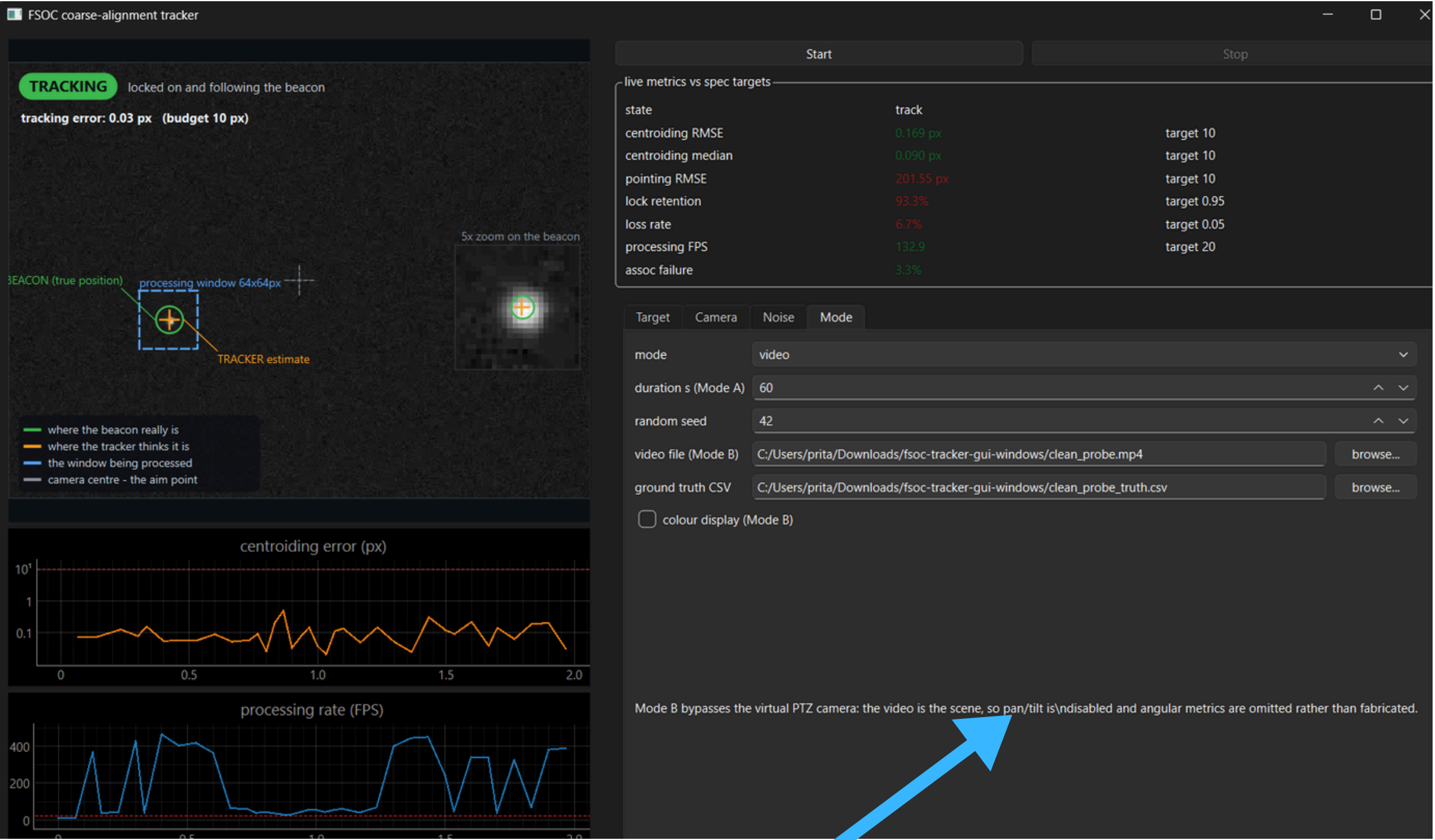
- Observe the surrounding environment,
- Acquire and detect the remote terminal or beacon,
- Estimate the position, and
- Continuously adjust the pointing direction to maintain visibility.

The participants shall develop this coarse alignment process in software, allowing to develop and validate tracking algorithms without specialized hardware and setup. The following section provides reference parameters and performance criteria to be considered for the software development.

Parameters and Specifications:

Functional Objective: Develop a software system that autonomously detects, identifies, and continuously tracks a designated moving target within a virtual scene by controlling a virtual camera viewport.

to test tracking algorithms hech tyancha main objective ahe. so basically aplyat jasa kalman ahe tashe i think ajun algos add karun tyancha benchmark havay tyana vegveglya simulative environments madhe



format this properly